

Assignment ::- 3

Name	Abhineet Tiwari
Roll No:	20965
Subject	DBMS

Q1) Create a database named CollegeDB and a table named Student.

The table should contain the following columns:

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
Age	INT	
Course	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	
Email	VARCHAR(60)	

Insert at least 10 records into the table. Some records should contain NULL values in the Email column.

```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

```
CREATE TABLE Student (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50),  
    Age INT,  
    Course VARCHAR(30),  
    City VARCHAR(30),  
    Marks INT,  
    Email VARCHAR(60)  
);
```

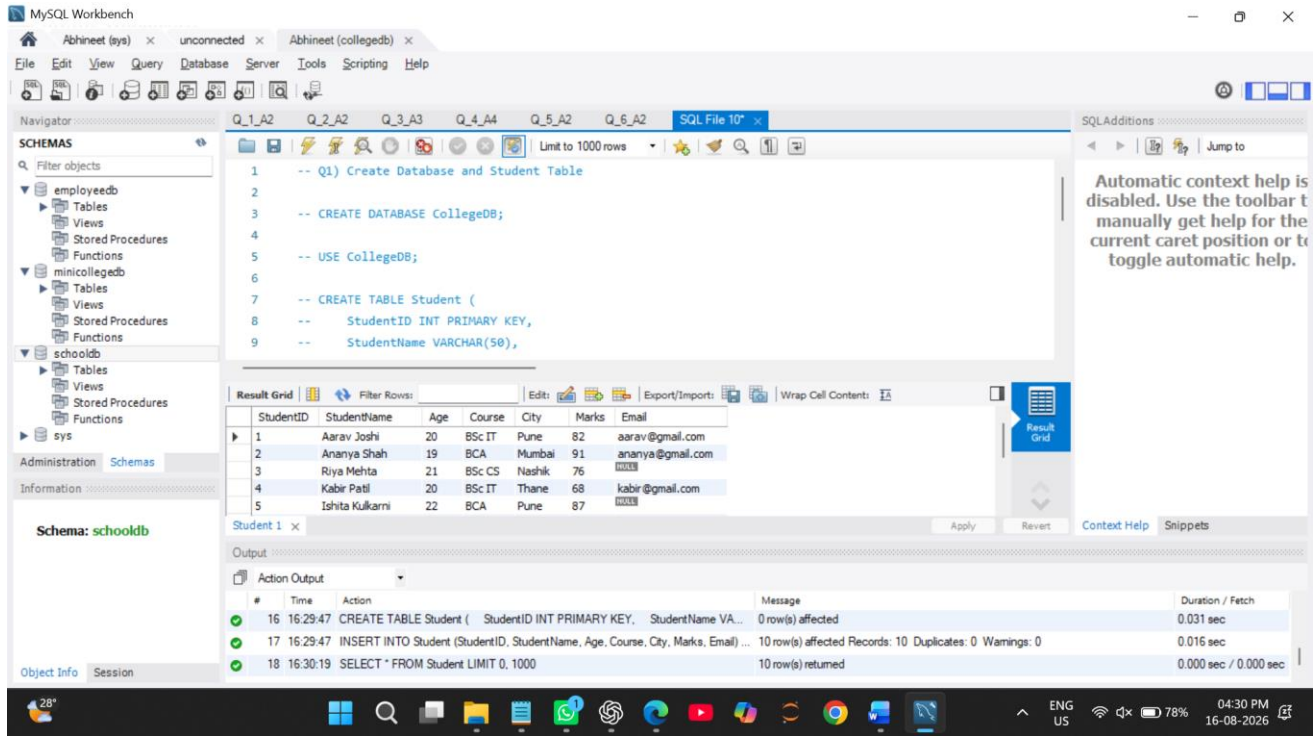
Insert 10 records

Some Email values are NULL

```
INSERT INTO Student  
(StudentID, StudentName, Age, Course, City, Marks, Email)  
VALUES  
(1, 'Aarav Joshi', 20, 'BSc IT', 'Pune', 82, 'aarav@gmail.com'),  
(2, 'Ananya Shah', 19, 'BCA', 'Mumbai', 91, 'ananya@gmail.com'),  
(3, 'Riya Mehta', 21, 'BSc CS', 'Nashik', 76, NULL),  
(4, 'Kabir Patil', 20, 'BSc IT', 'Thane', 68, 'kabir@gmail.com'),  
(5, 'Ishita Kulkarni', 22, 'BCA', 'Pune', 87, NULL),  
(6, 'Aditya More', 19, 'BSc CS', 'Mumbai', 73, 'aditya@gmail.com'),
```

(7, 'Snehal Deshmukh', 21, 'BSc IT', 'Nashik', 95, 'snehal@gmail.com'),
 (8, 'Rahul Naik', 20, 'BCA', 'Thane', 64, NULL),
 (9, 'Neha Pawar', 22, 'BSc CS', 'Pune', 89, 'neha@gmail.com'),
 (10, 'Omkar Jadhav', 21, 'BSc IT', 'Mumbai', 79, 'omkar@gmail.com');--

SELECT * FROM Student;



Q2) Write a MySQL query to display students who are from Mumbai and have marks greater than 80.

The output should include:

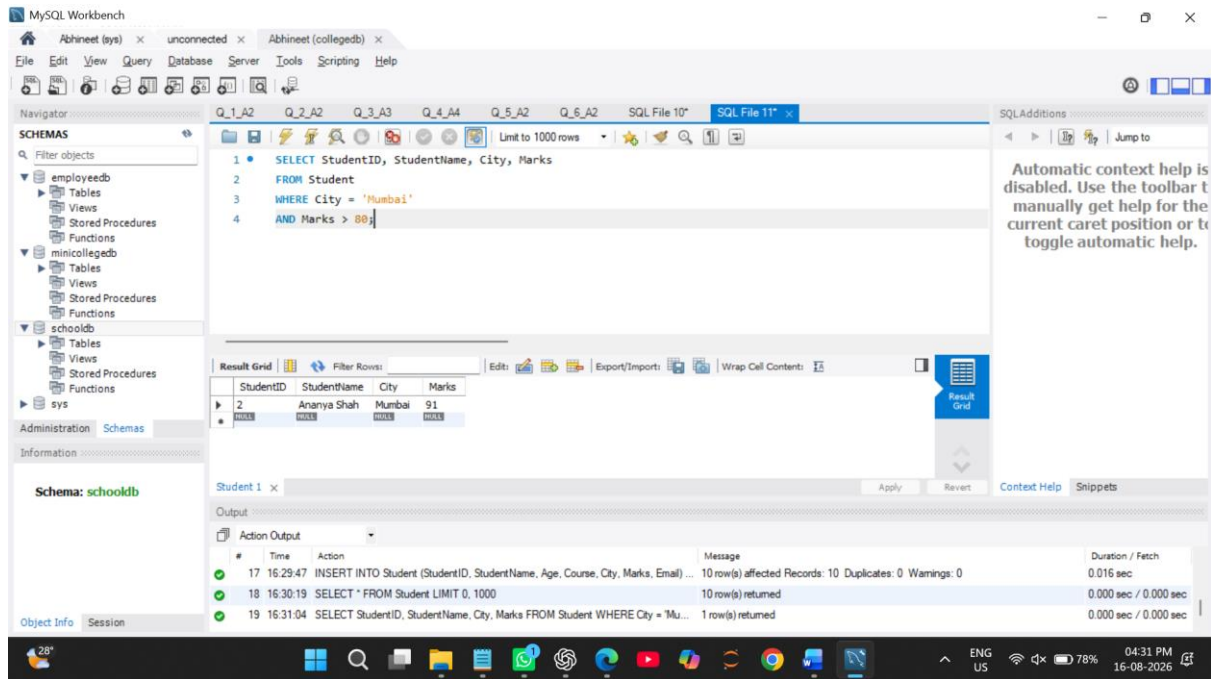
Column
StudentID
StudentName
City
Marks

SELECT StudentID, StudentName, City, Marks

FROM Student

WHERE City = 'Mumbai'

AND Marks > 80;



Q3) Write a MySQL query to display students who belong to either Pune or Nashik.

Use the OR operator and display the following column

Column
StudentName
Course
City

-- Use the OR operator and display StudentName, Course, and City.

`SELECT StudentName, Course, City`

`FROM Student`

`WHERE City = 'Pune'`

OR City = 'Nashik';

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
-- Q3) Write a MySQL query to display students who belong to either Pune or Nashik.
-- Use the OR operator and display StudentName, Course, and City.

SELECT StudentName, Course, City
FROM Student
WHERE City = 'Pune'
OR City = 'Nashik';
```

The Results window displays the following data:

StudentName	Course	City
Aarav Joshi	BSc IT	Pune
Riya Mehta	BSc CS	Nashik
Ishita Kulkarni	BCA	Pune
Snehal Deshmukh	BSc IT	Nashik
Neha Pawar	BSc CS	Pune

The Action Output window shows the execution of the query:

#	Time	Action	Message	Duration / Fetch
18	16:30:19	SELECT * FROM Student LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
19	16:31:04	SELECT StudentID, StudentName, City, Marks FROM Student WHERE City = 'Mu...	1 row(s) returned	0.000 sec / 0.000 sec
20	16:39:13	SELECT StudentName, Course, City FROM Student WHERE City = 'Pune' OR Ci...	5 row(s) returned	0.015 sec / 0.000 sec

Q4) Write a MySQL query to display students whose marks are between 70 and 90.

Use the BETWEEN operator and display:

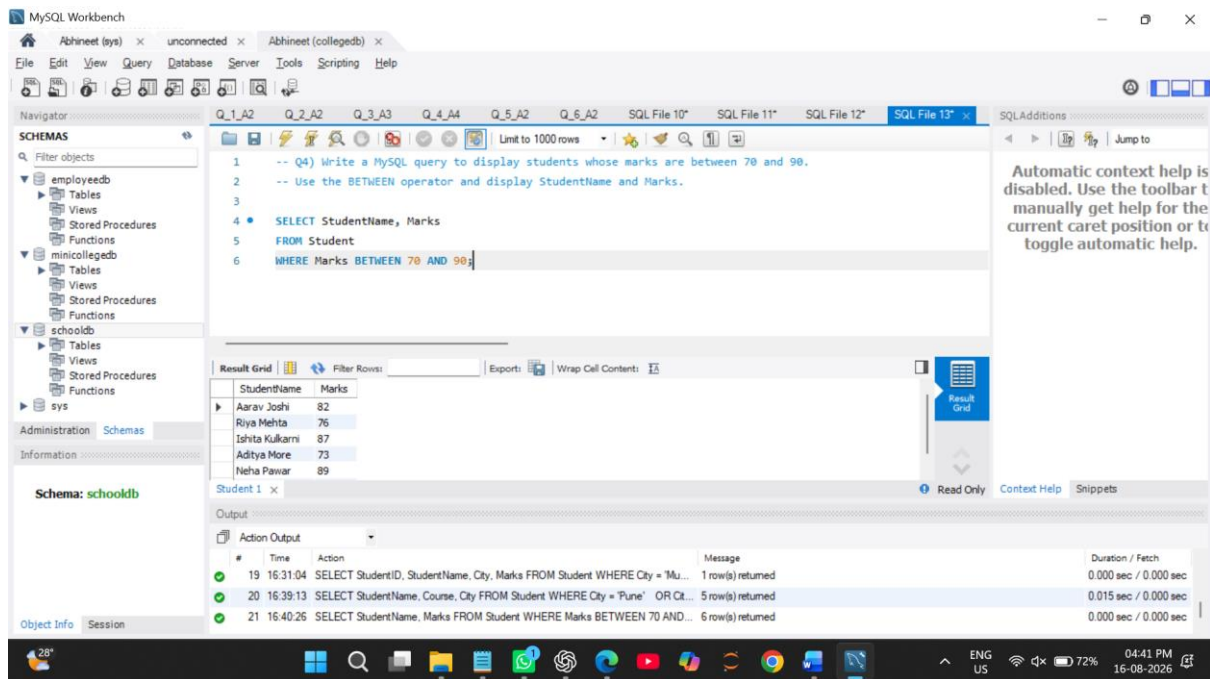
Column
StudentName
Marks

-- Use the BETWEEN operator and display StudentName and Marks.

SELECT StudentName, Marks

FROM Student

WHERE Marks BETWEEN 70 AND 90;



Q5) Write a MySQL query to display students whose names start with the letter A.

Use the LIKE operator and display:

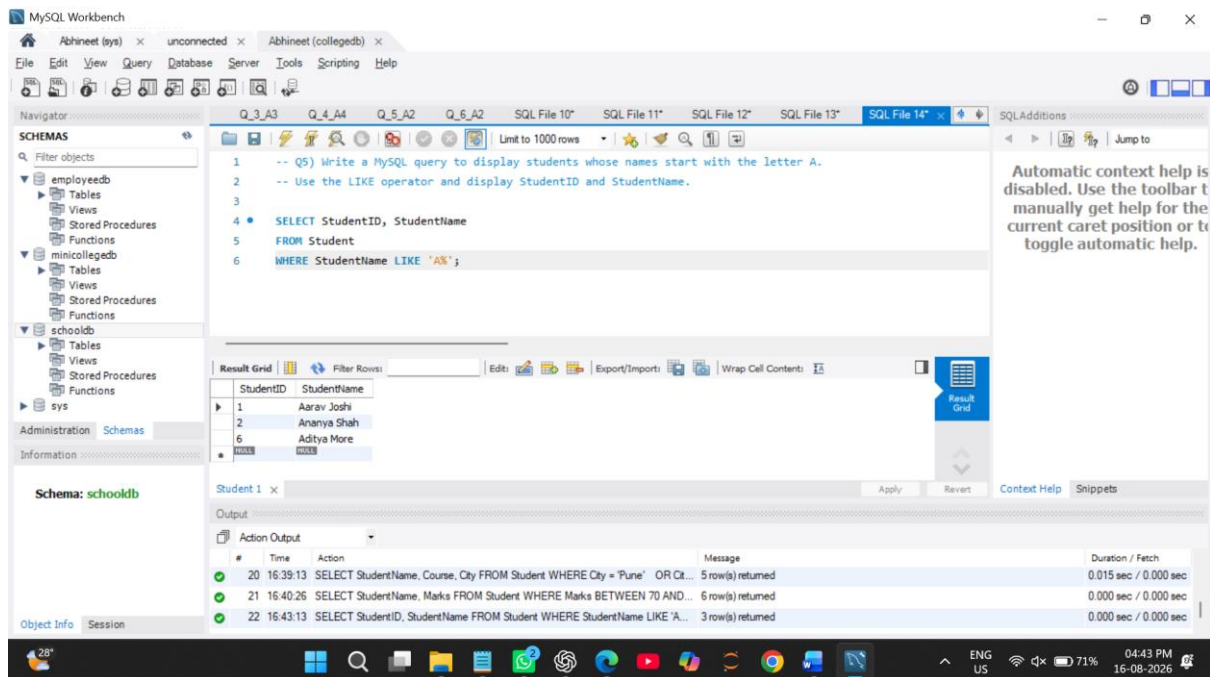
Column
StudentID
StudentName

-- Use the LIKE operator and display StudentID and StudentName.

SELECT StudentID, StudentName

FROM Student

WHERE StudentName LIKE 'A%';



Q6) Write a MySQL query to display students whose email address is missing.

Use the IS NULL operator and display:

Column
StudentID
StudentName
Email

-- Use the IS NULL operator and display StudentID, StudentName, and Email.

SELECT StudentID, StudentName, Email

FROM Student

WHERE Email IS NULL;

The screenshot shows the MySQL Workbench interface. The main editor displays a SQL query:

```
-- Q6) Write a MySQL query to display students whose email address is missing.
-- Use the IS NULL operator and display StudentID, StudentName, and Email.

SELECT StudentID, StudentName, Email
FROM Student
WHERE Email IS NULL;
```

The query is executed, and the results are shown in the Result Grid:

StudentID	StudentName	Email
3	Riya Mehta	NULL
5	Ishita Kulkarni	NULL
8	Rahul Naik	NULL

The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
21	16:40:26	SELECT StudentName, Marks FROM Student WHERE Marks BETWEEN 70 AND...	6 row(s) returned	0.000 sec / 0.000 sec
22	16:43:13	SELECT StudentID, StudentName FROM Student WHERE StudentName LIKE 'A...	3 row(s) returned	0.000 sec / 0.000 sec
23	16:45:13	SELECT StudentID, StudentName, Email FROM Student WHERE Email IS NULL L...	3 row(s) returned	0.000 sec / 0.000 sec

The Schema pane on the left shows the database structure, including the 'Student' table in the 'schooldb' schema. The right pane displays a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."